

HOSPITAL SMART CAR PARKING SYSTEM



ABSTRACT

This project focuses on designing a Smart Hospital Parking System using Verilog HDL. The system manages vehicle entry and exit by categorizing them into: Ambulance, Staff, and Patients.

It implements priority logic for ambulances and capacity tracking for other categories.

The design was modeled using a Finite State Machine (FSM) and verified through simulation to ensure efficient traffic flow and resource management.



INPUTS



clk, reset

System clock and reset signals



vehicle sensor

Detects if a car is at the entrance



ambulance sensor

High for emergency vehicles



pin valid

High for authorized staff PIN



exit staff, exit patient

Signals for vehicle exit



AMBULANCE



PRIORITY ACCESS

STAFF



CAPACITY TRACKING

PATIENTS



CAPACITY TRACKING

OUTPUTS



gate open

Signal to open the parking gate



ambulance pass

Priority access indicator



staff full, patient full

Capacity limit indicators



staff vacant, patient vacant

Spot availability indicators



staff count, patient count

4-bit status for current occupancy